

The Nature of Volcanoes

Volcanoes are mountains but they are very different from other mountains; they are not formed by folding and crumpling or by uplift and erosion. Instead, volcanoes are built by the accumulation of their own eruptive products -- lava, bombs (crusted over ash flows, and tephra (airborne ash and dust). A volcano is most commonly a conical hill or mountain built around a vent that connects with reservoirs of molten rock below the surface of the Earth. The term *volcano* also refers to the opening or vent through which the molten rock and associated gases are expelled.

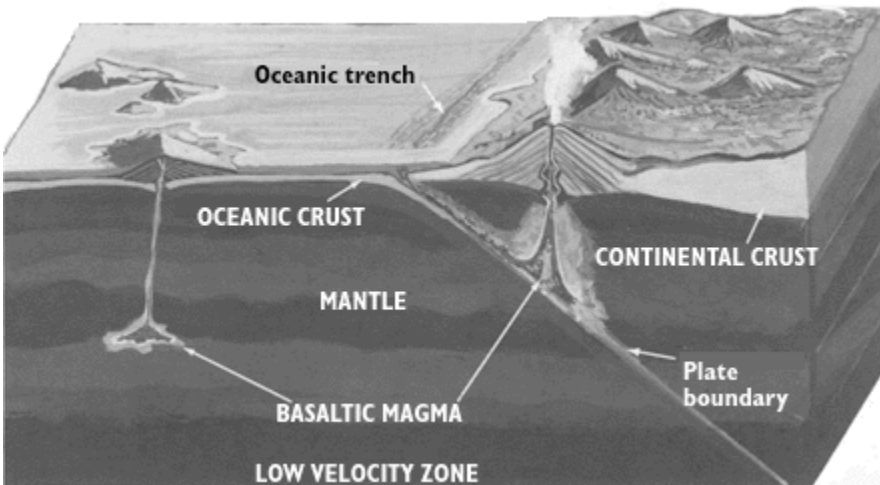
Driven by buoyancy and gas pressure the molten rock, which is lighter than the surrounding solid rock forces its way upward and may ultimately break though zones of weaknesses in the Earth's crust. If so, an eruption begins, and the molten rock may pour from the vent as non-explosive lava flows, or it may shoot violently into the air as dense clouds of lava fragments. Larger fragments fall back around the vent, and accumulations of fall-back fragments may move downslope as ash flows under the force of gravity. Some of the finer ejected materials may be carried by the wind only to fall to the ground many miles away. The finest ash particles may be injected miles into the atmosphere and carried many times around the world by stratospheric winds before settling out.



Fountaining lava and volcanic debris during the 1959 Kilauea Iki eruption of Kilauea Volcano, Hawaii.

Molten rock below the surface of the Earth that rises in volcanic vents is known as *magma*, but after it erupts from a volcano it is called *lava*. Originating many tens of miles beneath the ground, the ascending magma commonly contains some crystals, fragments of surrounding (unmelted) rocks, and dissolved gases, but it is primarily a liquid composed principally of oxygen, silicon, aluminum, iron, magnesium, calcium, sodium, potassium, titanium, and manganese. Magmas also contain many other chemical elements in trace quantities. Upon cooling, the liquid magma may precipitate crystals of various minerals until solidification is complete to form an *igneous* or *magmatic* rock.

The diagram below shows that heat concentrated in the Earth's upper *mantle* raises temperatures sufficiently to melt the rock locally by fusing the materials with the lowest melting temperatures, resulting in small, isolated blobs of magma. These blobs then collect, rise through conduits and fractures, and some ultimately may re-collect in larger pockets or reservoirs ("holding tanks") a few miles beneath the Earth's surface. Mounting pressure within the reservoir may drive the magma further upward through structurally weak zones to erupt as lava at the surface. In a continental environment, magmas are generated in the Earth's crust as well as at varying depths in the upper mantle. The variety of molten rocks in the crust, plus the possibility of mixing with molten materials from the underlying mantle, leads to the production of magmas with widely different chemical compositions.



An idealized diagram of a volcano in an oceanic environment (left) and in a continental environment (right).

If magmas cool rapidly, as might be expected near or on the Earth's surface, they solidify to form igneous rocks that are finely crystalline or glassy with few crystals. Accordingly, lavas, which of course are very rapidly cooled, form volcanic rocks typically characterized by a small percentage of crystals or fragments set in a matrix of *glass* (quenched or super-cooled magma) or finer grained crystalline materials. If magmas never breach the surface to erupt and remain deep underground, they cool much more slowly and thus allow ample time to sustain crystal precipitation and growth, resulting in the formation of coarser grained, nearly completely crystalline, igneous rocks. Subsequent to final crystallization and solidification, such rocks can be exhumed by erosion many thousands or millions of years later and be exposed as large bodies of so-called *granitic* rocks, as, for example, those spectacularly displayed in Yosemite National Park and other parts of the majestic Sierra Nevada mountains of California.



Two Polynesian terms are used to identify the surface character of Hawaiian lava flows. *Aa*, a basalt with a rough, blocky appearance, much like furnace slag, is shown at the left. *Pahoehoe*, a more fluid variety with a smooth, satiny and sometimes glassy appearance, is shown at the right.

Lava is red hot when it pours or blasts out of a vent but soon changes to dark red, gray, black, or some other color as it cools and solidifies. Very hot, gas-rich lava containing abundant iron and magnesium is fluid and flows like hot tar, whereas cooler, gas-poor lava high in silicon, sodium, and potassium flows sluggishly, like thick honey in some cases or in others like pasty, blocky masses.

All magmas contain dissolved gases, and as they rise to the surface to erupt, the confining pressures are reduced and the dissolved gases are liberated either quietly or explosively. If the lava is a thin fluid (not viscous), the gases may escape easily. But if the lava is thick and pasty (highly viscous), the gases will not move freely but will build up tremendous pressure, and ultimately escape with explosive violence. Gases in lava may be compared with the gas in a bottle of a carbonated soft drink. If you put your thumb over the top of the bottle and shake it vigorously, the gas separates from the drink and forms bubbles. When you remove your thumb abruptly, there is a miniature explosion of gas and liquid. The gases in lava behave in somewhat the same way. Their sudden expansion causes the terrible explosions that throw out great masses of solid rock as well as lava, dust, and ashes.

The violent separation of gas from lava may produce rock froth called *pumice*. Some of this froth is so light--because of the many gas bubbles--that it floats on water. In many eruptions, the froth is shattered explosively into small fragments that are hurled high into the air in the form of volcanic cinders (red or black), volcanic ash (commonly tan or gray), and volcanic dust.



During the 1959 eruption of Kilauea Iki, fountaining lava and volcanic debris completely blocked several of the roads in the Hawaii Volcanoes National Park.